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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 5.1 Introduction to the IRIS data set
5  Machine Learning for Engineering Problem Solving
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7  """
8
9  import IPython as IP
10 IP.get_ipython().run_line_magic('reset', '-sf')
11
12 import matplotlib.pyplot as plt
13 import sklearn as sk
14
15 cc = plt.rcParams['axes.prop_cycle'].by_key()['color']
16 plt.close('all')
17
18
19 %% Load your data
20
21 # We will use the Iris data set. This dataset was created by biologist Ronald
22 # Fisher in his 1936 paper "The use of multiple measurements in taxonomic
23 # problems" as an example of linear discriminant analysis
24
25 iris = sk.datasets.load_iris()
26
27 # for simplicity, extract some of the data sets
28 X = iris['data'] # this contains the length of the pedals and sepals
29 Y = iris['target'] # contains what type of flower it is
30 Y_names = iris['target_names'] # contains the name that aligns with the type of the
31 flower
32 feature_names = iris['feature_names'] # the names of the features
33
34 # plot the Sepal data
35 plt.figure(figsize=(6.5,3))
36 plt.subplot(121)
37 plt.grid(True)
38 plt.scatter(X[Y==0,0],X[Y==0,1],marker='o')
39 plt.scatter(X[Y==1,0],X[Y==1,1],marker='s')
40 plt.scatter(X[Y==2,0],X[Y==2,1],marker='d')
41 plt.xlabel(feature_names[0])
42 plt.ylabel(feature_names[1])
43
44 plt.subplot(122)
45 plt.grid(True)
46 plt.scatter(X[Y==0,2],X[Y==0,3],marker='o',label=Y_names[0])
47 plt.scatter(X[Y==1,2],X[Y==1,3],marker='s',label=Y_names[1])
48 plt.scatter(X[Y==2,2],X[Y==2,3],marker='d',label=Y_names[2])
49 plt.xlabel(feature_names[2])
50 plt.ylabel(feature_names[3])
51 plt.legend(framealpha=1)
52 plt.tight_layout()
53

```